6. Supplementary Material

In this supplementary material, we provide the full results of our complete unsupervised domain adaptation (UDA) pipeline for each evaluated clinically relevant cross-domain scenario in Section 6.1. In addition, we provide the full within-algorithm validator reliability analyses, reported per algorithm and per scenario, in Section 6.2.

6.1. Full Results on the Complete UDA Pipeline

This section provides the full per-scenario complete UDA results. Each table reports, for one cross-domain scenario, the target accuracy of the checkpoint selected by every validator (rows) for every algorithm (columns), along with *Oracle*, *SourceOnly*, and *TargetOnly* references. For each scenario, the first table reports mean±std and the immediately following table reports median with 95% CI (e.g., Table 5 is mean±std and Table 6 is median with 95% CI, and analogously for the remaining ten scenarios). Tables 5–25 cover the eleven clinically relevant transfer scenarios. Brain MRI: ADNI-1→ADNI-2 (Table 5), ADNI-1→ADNI-3 (Table 7), ADNI-2→ADNI-1 (Table 9), ADNI-2→ADNI-3 (Table 11), ADNI-1+2→AIBL (Table 13). Chest X-ray: RSNA→Child CXR (Table 15), Child CXR→RSNA (Table 17), LDD→CRD (Table 19), CRD→LDD (Table 21). Retinal: OCT→SLO (Table 23), SLO→OCT (Table 25).

We also assess whether adaptation works in principle and the size of the selection gap using the median with 95% confidence intervals, shown in Figures 8 and 9. The conclusions match those drawn from the mean ± standard deviation in the main paper.

6.2. Full Validator Reliability Analyses for Within-Algorithm Selection

This section provides the full per-scenario validator reliability analyses. Each table reports, for one cross-domain scenario, the within-algorithm Spearman correlation (ρ) between each validation score and target accuracy, computed over the checkpoints of each algorithm. Positive values always indicate ranking in the intended direction. The ρ values here are means across folds or random seeds, whereas the example scatter plots of the main paper show a single fold or random seed for readability, so individual cells may not match exactly. Tables 27–37 cover the eleven clinically relevant transfer scenarios. Brain MRI: ADNI-1→ADNI-2 (Table 27), ADNI-1→ADNI-3 (Table 28), ADNI-2→ADNI-1 (Table 29), ADNI-2→ADNI-3 (Table 30), ADNI-1+2→AIBL (Table 31). Chest X-ray: RSNA→Child CXR (Table 32), Child CXR→RSNA (Table 33), LDD→CRD (Table 34), CRD→LDD (Table 35). Retinal: OCT→SLO (Table 36), SLO→OCT (Table 37).

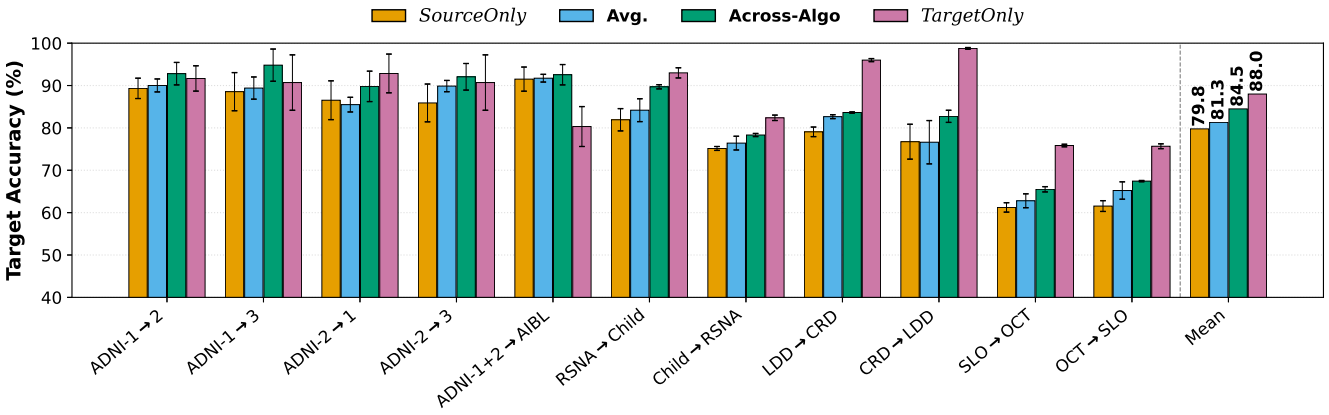


Figure 8. Target accuracy under *Oracle* selection across UDA scenarios, reported as the median with 95% confidence intervals. Bars show *SourceOnly* (no adaptation), the per-algorithm average (**Avg.**), **Across-Algo** (selection over all algorithms' pooled checkpoints), and *TargetOnly* (labeled-target). **Across-Algo** exceeds *SourceOnly* in all scenarios and approaches *TargetOnly*, indicating a capable adapted model usually exists. Error bars: 95% confidence interval.

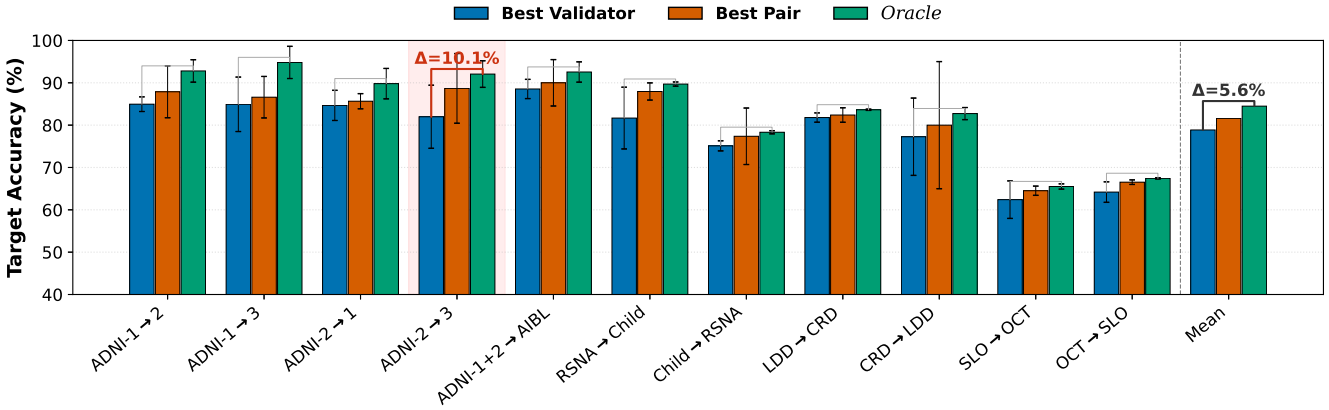


Figure 9. Target accuracy across UDA scenarios under three selections (%), reported as the median with 95% confidence intervals. Bars show the accuracy of the model selected by the best validator over the across-algorithm pool (**Best Validator**), by the best algorithm–validator combination (**Best Pair**), and by the across-algorithm *Oracle*. The first two fall below the *Oracle* in every scenario, with the gap up to 10.1 points and averaging 5.6 (**Mean**). **Best Validator** and **Best Pair** differ from scenario to scenario and are identified using target labels, so they cannot be chosen in advance; the gap realized at deployment can only be larger. Error bars: 95% confidence interval.

Table 5. Complete UDA results on ADNI-1→ADNI-2 (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	88.2±2.0	89.3±4.0	90.0±4.0	89.9±3.0	91.0±3.5	91.0±1.7	89.9±4.4	85.7±9.8	87.4±4.3	92.0±0.93	89.5±2.0	93.2±2.0	90.9±2.4
Source-Risk	81.2±5.8	82.1±3.1	81.8±4.0	81.1±8.0	81.1±9.0	83.4±3.3	84.1±4.1	73.2±12	81.5±6.3	86.8±2.8	-	84.4±2.5	-
IWCV	76.8±9.7	80.3±5.3	74.6±12	81.1±6.4	80.1±8.4	76.0±10	76.4±3.8	72.3±11	80.1±11	84.3±2.8	-	76.7±4.4	-
DEV	78.0±11	82.2±3.2	84.5±6.2	82.0±6.7	84.9±6.6	76.9±11	75.9±7.3	70.3±15	68.6±19	81.4±8.2	-	79.4±13	-
DEV-N	80.9±5.8	81.7±2.8	84.9±2.0	81.8±8.4	80.8±9.1	83.5±3.3	82.9±3.4	76.0±12	81.0±6.2	87.6±1.9	-	85.1±1.3	-
Entropy	60.9±15	58.8±20	56.7±15	57.0±16	50.0±15	60.2±15	57.5±16	57.9±13	70.1±19	56.9±15	-	50.0±10	-
InfoMax	84.2±2.1	84.8±5.7	87.3±4.4	86.0±1.5	88.2±3.8	81.8±9.6	83.5±6.9	78.9±10	83.3±5.6	86.3±2.2	-	83.2±7.2	-
Corr-C	84.6±4.1	80.4±4.3	83.4±3.6	83.1±3.6	84.2±6.7	77.8±5.3	80.4±3.4	78.3±8.8	84.1±4.4	85.6±5.1	-	79.6±3.0	-
MCC (V)	84.2±2.7	84.4±6.4	84.9±3.3	84.5±4.0	86.4±3.7	80.4±8.3	85.9±3.6	80.7±13	83.1±3.7	87.1±3.7	-	80.4±2.3	-
BNM (V)	84.7±2.1	84.8±5.7	87.5±4.0	84.9±3.8	87.2±2.2	81.8±9.6	83.5±6.9	78.9±10	83.3±5.6	86.3±2.2	-	81.8±9.6	-
ClassAMI	80.9±6.4	60.0±7.3	82.1±3.1	82.7±3.8	84.6±3.5	83.3±2.4	79.8±2.0	79.7±8.2	59.2±9.8	81.6±2.1	-	81.0±2.3	-
SND	73.9±12	83.4±4.1	80.0±2.6	79.2±5.8	85.5±4.9	78.2±5.9	78.5±8.6	75.8±12	82.2±5.6	80.7±8.7	-	80.7±8.7	-
MixVal	75.9±12	57.4±6.4	77.9±8.7	69.8±13	66.8±16	82.1±8.7	71.3±16	70.2±15	71.8±16	60.0±16	-	61.1±18	-
TransScore	85.2±2.8	82.7±4.8	86.9±2.2	85.9±2.0	79.3±14	81.4±8.2	76.4±9.7	76.9±12	83.3±5.4	87.8±3.1	-	78.5±11	-

Table 6. Complete UDA results on ADNI-1→ADNI-2 (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 5. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	AD2A	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	89.3±2.4	88.6±5.4	89.1±5.2	89.8±3.7	91.7±4.8	91.3±1.9	90.3±4.8	92.3±1.2	89.3±12	87.7±5.4	90.0±1.5	92.8±2.6	91.7±3.0
Source-Risk	82.4±7.9	82.2±3.8	82.0±4.8	82.7±10.0	82.0±12	84.4±4.1	81.4±4.1	86.7±3.1	74.6±14	81.1±7.5	-	84.4±3.0	-
IWCV	75.8±13	80.7±7.0	70.7±14	80.7±8.4	83.7±10	79.2±12	76.5±4.8	86.9±7.5	74.6±14	85.8±13	-	79.2±5.2	-
DEV	83.3±13	82.2±3.8	83.7±8.1	83.7±8.4	87.5±7.0	80.8±13	79.4±8.9	81.6±9.2	66.0±15	69.7±21	-	81.6±17	-
Entropy	53.2±17	50.0±0.85	50.0±17	50.0±18	50.0±0.0	50.0±17	50.0±18	50.0±17	50.0±19	77.3±19	-	50.0±0.0	-
InfoMax	85.0±2.5	82.3±7.2	86.5±6.0	86.8±1.9	86.7±4.7	83.3±12	84.3±8.5	85.7±2.7	84.5±12	81.1±7.4	-	83.3±9.0	-
Corr-C	85.0±4.5	82.5±4.9	82.3±4.3	82.7±4.7	87.3±7.7	79.3±6.5	82.0±4.2	87.9±6.1	81.7±11	81.4±5.0	-	79.3±3.8	-
MCC (V)	84.8±3.7	84.2±8.6	84.5±4.3	85.6±4.9	86.7±4.6	83.3±11	86.7±4.5	87.5±4.7	85.2±17	81.4±4.7	-	83.3±11	-
BNM (V)	85.7±2.5	82.3±7.2	86.5±5.2	86.8±4.6	86.7±2.4	83.3±12	84.3±8.5	85.7±2.7	84.5±12	81.1±7.4	-	83.3±12	-
SND	79.0±14	83.3±4.8	79.7±3.6	75.9±6.8	84.5±6.3	78.8±8.0	79.6±11	82.8±5.4	81.7±14	83.4±7.6	-	79.7±10	-
ClassAMI	82.5±8.0	62.1±9.6	81.1±4.1	83.0±4.9	83.9±4.5	83.3±3.3	79.4±2.6	81.1±2.7	84.4±9.7	58.5±12	-	81.1±2.8	-
DEV-N	80.7±7.9	82.2±3.4	85.2±2.8	82.7±10.0	82.0±12	84.4±4.1	81.4±4.1	86.9±2.4	74.6±14	78.8±7.1	-	85.0±1.7	-
MixVal	80.3±15	55.7±8.5	78.0±11	71.3±16	65.2±18	83.2±10	77.7±18	68.8±20	67.6±19	79.8±20	-	53.0±19	-
TransScore	85.7±3.6	82.4±6.7	86.9±2.7	86.4±2.5	85.8±17	83.3±11	79.9±12	87.7±3.5	81.9±15	81.1±7.0	-	83.2±15	-

Table 7. Complete UDA results on ADNI-1→ADNI-3 (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	90.2±3.7	87.3±4.5	90.7±3.1	93.5±3.5	90.8±4.1	90.8±3.3	89.0±5.6	86.9±5.8	88.6±3.5	91.1±4.3	89.9±2.0	94.2±3.1	90.2±4.6
Source-Risk	82.8±8.3	78.0±9.9	83.8±5.3	82.7±8.8	82.4±7.2	77.6±6.2	78.0±7.1	77.9±10	82.9±5.1	83.0±6.7	-	83.5±4.8	-
IWCV	70.0±10	76.7±11	79.6±4.9	84.1±5.9	74.7±15	74.0±12	56.5±9.1	69.3±11	79.2±6.7	75.3±12	-	64.9±15	-
DEV	73.9±12	69.4±11	82.2±4.9	86.5±5.0	81.1±7.2	79.9±6.5	62.9±13	72.8±11	79.4±7.1	81.6±5.8	-	84.0±7.5	-
DEV-N	83.3±7.1	79.2±9.7	83.8±5.3	82.7±8.8	82.4±5.0	77.3±5.7	76.1±8.5	76.6±12	75.2±9.8	83.2±4.4	-	84.5±5.0	-
Entropy	67.8±13	66.8±15	57.2±15	57.3±14	63.2±18	66.1±17	64.0±11	64.1±18	62.5±17	50.3±0.72	-	50.0±0.0	-
InfoMax	79.0±5.9	72.0±14	79.7±5.0	78.2±4.1	79.4±6.8	70.6±7.9	72.7±6.5	72.4±12	80.2±5.8	80.2±5.7	-	70.6±7.9	-
Corr-C	79.7±6.7	70.8±5.1	76.8±1.5	75.9±3.7	74.5±4.3	71.3±8.3	69.9±5.0	67.1±13	76.1±4.8	72.1±11	-	68.7±8.5	-
MCC (V)	79.1±7.9	74.9±14	84.6±2.3	84.5±4.0	79.9±6.3	75.9±9.8	72.4±4.2	73.8±18	85.1±3.9	83.5±7.6	-	75.9±9.8	-
BNM (V)	78.2±7.3	75.0±15	79.7±5.0	78.2±4.1	79.4±6.8	70.6±7.9	72.7±6.5	74.3±13	80.9±4.9	80.2±5.7	-	70.6±7.9	-
ClassAMI	80.7±6.5	54.6±18	83.6±2.0	82.9±2.2	80.6±6.7	81.6±5.3	81.3±7.1	74.2±9.8	58.0±11	80.1±7.3	-	80.6±3.6	-
SND	73.8±6.0	71.6±5.0	77.5±4.5	75.4±6.0	76.8±4.3	73.2±5.3	71.2±9.0	69.4±8.8	77.4±3.2	74.5±2.0	-	76.2±3.9	-
MixVal	78.2±11	64.9±18	73.3±9.1	78.1±12	66.7±14	82.3±4.0	72.6±14	72.7±11	53.4±6.0	59.8±12	-	64.7±13	-
TransScore	76.3±6.4	69.9±13	81.7±2.7	81.1±5.9	76.9±5.0	69.6±7.1	47.3±19	73.9±13	75.0±6.0	72.9±22	-	62.6±16	-

Table 8. Complete UDA results on ADNI-1→ADNI-3 (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 7. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	AD2A	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	88.5±4.5	86.2±5.1	90.0±3.5	93.8±4.7	91.6±5.5	90.7±4.4	87.8±6.9	90.8±5.0	86.0±7.4	87.6±4.6	89.4±2.6	94.8±3.8	90.7±6.5
Source-Risk	82.8±9.7	79.2±10	85.4±6.7	80.1±11	82.3±9.8	78.5±8.3	79.1±8.7	84.7±8.2	80.5±13	84.9±6.3	-	82.2±6.4	-
IWCV	64.3±11	78.9±15	81.5±5.3	86.1±7.6	77.8±19	79.4±15	56.0±11	75.5±17	69.8±14	76.1±7.6	-	60.5±16	-
DEV	73.1±14	73.6±14	81.8±6.7	86.1±6.5	79.5±9.7	83.0±7.8	60.5±17	81.0±7.8	74.7±14	76.1±8.1	-	84.1±9.7	-
Entropy	68.4±17	64.3±19	50.0±17	50.0±16	51.2±20	64.5±19	69.6±13	50.0±0.81	66.7±23	51.0±17	-	50.0±0.0	-
InfoMax	77.0±7.3	73.3±18	80.8±6.6	79.7±5.0	77.4±8.7	72.4±11	75.0±8.2	79.3±6.8	76.5±14	78.4±6.9	-	72.4±11	-
Corr-C	79.0±8.0	72.6±6.3	76.7±1.8	76.7±4.0	76.0±5.2	72.4±11	70.7±6.3	74.6±14	72.6±14	77.9±6.1	-	71.6±11	-
MCC (V)	77.0±10	80.7±18	84.2±3.1	86.0±4.5	76.8±7.4	72.2±12	72.8±5.3	79.3±7.9	78.1±23	86.6±4.9	-	72.2±12	-
BNM (V)	75.5±8.9	82.4±18	80.8±6.6	79.7±5.0	77.4±8.7	72.4±11	75.0±8.2	79.3±6.8	78.1±17	78.4±5.2	-	72.4±11	-
SND	74.2±7.9	72.6±6.2	78.7±5.5	74.3±7.5	77.1±5.4	74.9±6.7	69.3±11	74.0±2.3	72.8±11	77.9±4.1	-	77.1±5.2	-
ClassAMI	84.2±7.6	53.2±23	84.4±2.2	82.2±2.5	78.5±8.6	80.5±6.3	78.2±7.5	79.2±7.8	77.5±13	55.6±14	-	80.2±4.2	-
DEV-N	82.8±7.8	77.6±11	85.4±6.7	80.1±11	80.8±6.4	78.5±7.6	78.0±11	84.7±4.5	80.5±16	77.9±11	-	84.9±6.4	-
MixVal	80.9±14	57.7±18	70.8±11	80.7±17	69.4±16	81.6±4.9	75.4±17	54.8±15	74.8±15	56.1±7.2	-	65.3±16	-
TransScore	74.3±8.4	71.7±18	81.1±3.0	82.5±8.0	77.4±6.7	71.9±9.2	40.1±23	80.1±26	78.1±16	77.9±6.4	-	70.9±20	-

Table 9. Complete UDA results on ADNI-2→ADNI-1 (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	85.5±3.9	86.6±3.1	86.6±3.7	87.4±2.8	87.2±3.2	85.5±3.8	86.9±3.5	85.0±4.1	85.2±5.3	87.2±3.1	86.4±0.88	89.5±2.6	91.0±4.2
Source-Risk	77.4±5.0	80.4±4.7	78.8±6.9	75.2±7.5	79.8±5.4	80.2±6.3	78.7±5.0	76.7±4.3	77.5±8.4	80.8±4.7	-	78.5±4.7	-
IWCV	75.1±3.6	79.5±3.5	79.0±5.2	79.2±7.3	76.5±6.0	77.7±6.1	80.8±2.4	75.3±7.7	70.0±9.8	79.5±6.1	-	81.9±4.4	-
DEV	76.3±2.8	79.5±3.5	80.7±5.6	77.8±8.4	81.2±3.4	77.7±6.1	78.1±3.2	73.8±5.7	69.6±10.0	80.5±4.1	-	78.1±3.5	-
DEV-N	78.8±3.1	79.7±5.0	79.4±6.8	78.2±6.5	78.9±5.9	82.8±3.9	80.5±3.4	77.0±4.5	76.8±9.0	84.8±1.5	-	84.4±2.8	-
Entropy	53.9±6.1	56.6±14	61.7±16	64.2±19	50.5±1.2	67.5±18	68.7±17	66.0±15	70.0±14	56.9±16	-	56.1±14	-
InfoMax	81.0±4.3	79.1±6.9	83.2±3.5	80.7±4.4	81.5±2.2	82.0±4.4	79.5±6.0	79.7±6.4	76.3±8.2	84.1±5.5	-	82.0±4.4	-
Corr-C	79.6±5.5	80.3±4.5	81.2±5.4	80.0±5.1	81.5±3.5	80.8±5.9	77.2±5.0	80.4±5.2	77.0±8.4	83.8±5.4	-	80.8±5.9	-
MCC (V)	80.5±4.5	81.3±4.7	82.2±3.9	82.4±3.9	82.5±3.5	73.6±11	78.5±4.5	80.1±6.3	77.4±7.2	84.1±3.7	-	73.6±11	-
BNM (V)	80.9±4.1	79.3±6.8	83.2±3.5	80.7±4.4	82.0±3.1	82.0±4.4	80.2±6.3	80.5±6.6	76.3±8.2	83.8±5.5	-	82.0±4.4	-
ClassAMI	76.3±3.1	77.7±2.0	81.5±4.0	80.5±4.1	80.8±3.8	80.4±3.9	78.2±5.3	78.1±5.4	64.4±15	79.0±5.9	-	80.0±4.2	-
SND	81.9±7.2	77.5±8.0	75.7±1.9	77.4±6.1	78.8±4.6	67.5±7.8	75.8±8.0	75.0±3.1	74.6±3.6	72.3±1.4	-	75.9±8.7	-
MixVal	83.2±6.5	73.9±12	71.0±13	82.3±4.0	72.7±17	73.4±14	78.9±7.8	75.0±13	60.8±16	68.8±14	-	63.3±11	-
TransScore	78.3±5.6	79.4±3.5	82.4±3.2	80.5±4.7	81.8±3.3	77.2±3.5	79.1±6.5	77.8±4.0	77.2±7.0	82.2±3.4	-	78.4±5.0	-

Table 10. Complete UDA results on ADNI-2→ADNI-1 (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 9. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	AD2A	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	86.5±4.6	85.4±3.7	85.9±5.0	87.5±3.7	87.0±4.6	84.6±4.8	87.0±3.9	85.7±3.4	84.1±5.2	82.0±5.7	85.5±1.7	89.8±3.6	92.8±4.6
Source-Risk	76.9±6.8	81.2±6.3	75.1±8.2	76.2±9.9	78.7±6.9	80.2±7.6	77.5±6.0	81.1±5.7	78.5±5.1	72.6±9.2	-	78.0±5.7	-
IWCV	75.5±5.0	78.6±4.5	76.0±5.7	78.0±9.7	76.5±7.5	75.5±8.0	80.2±3.0	79.9±8.3	74.3±8.9	66.9±13	-	80.2±5.3	-
DEV	76.1±3.6	78.6±4.5	80.0±7.0	77.5±12	81.8±3.9	75.5±8.0	78.9±4.1	79.4±5.2	74.9±6.4	66.5±13	-	78.5±4.5	-
Entropy	50.0±7.0	50.0±16	50.0±15	52.3±19	50.0±1.3	67.2±20	79.4±16	50.0±17	73.8±14	70.6±19	-	50.0±15	-
InfoMax	81.0±4.9	78.5±9.0	83.3±4.8	80.2±5.3	82.0±2.8	79.9±5.2	77.5±7.9	84.7±7.0	77.5±8.3	75.3±11	-	79.9±5.2	-
Corr-C	77.4±6.7	78.3±5.2	84.5±6.2	78.6±6.2	81.0±4.5	79.8±8.1	75.7±6.0	84.5±6.7	78.6±6.3	76.2±12	-	79.8±8.1	-
MCC (V)	82.0±5.7	81.8±6.2	80.4±4.8	80.4±4.8	81.6±4.4	67.2±12	79.4±5.6	83.2±5.0	78.8±8.3	75.3±9.3	-	67.2±12	-
BNM (V)	81.0±4.7	78.5±9.0	83.3±4.8	80.2±5.3	82.0±4.0	79.9±5.2	77.5±7.9	83.2±7.0	77.5±8.3	75.3±11	-	79.9±5.2	-
SND	83.1±9.6	77.6±11	76.2±2.2	79.8±7.5	78.8±5.7	65.5±10	72.0±8.2	72.5±1.8	75.2±4.0	75.1±4.7	-	79.4±10	-
ClassAMI	76.9±3.6	78.6±2.5	80.7±5.4	80.3±5.3	81.6±5.1	79.8±5.4	76.4±6.3	79.6±7.2	78.1±7.2	64.3±20	-	79.3±5.7	-
DEV-N	77.4±3.7	79.8±6.3	75.8±8.1	77.6±8.8	77.1±7.2	83.3±5.0	80.0±3.7	85.7±1.8	78.8±5.3	70.8±9.5	-	84.7±3.6	-
MixVal	85.3±8.1	71.3±15	75.0±13	81.1±5.0	82.2±17	79.2±15	81.1±10	70.0±14	78.8±17	53.7±20	-	58.1±13	-
TransScore	77.1±6.5	78.5±4.4	84.4±3.4	79.8±5.8	80.9±4.5	77.4±4.4	77.5±8.5	82.2±4.4	79.7±5.0	78.5±9.6	-	78.6±6.8	-

Table 11. Complete UDA results on ADNI-2→ADNI-3 (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	88.1±3.8	89.3±3.8	91.1±3.3	91.2±3.0	90.7±2.9	90.1±4.1	90.1±2.9	89.2±3.1	90.0±3.0	91.2±2.8	90.3±0.74	92.1±2.4	90.2±4.6
Source-Risk	82.3±7.3	81.2±5.7	80.8±6.8	84.9±5.9	82.5±2.8	82.6±6.4	79.2±7.4	81.3±5.4	83.3±6.0	85.2±6.4	-	82.7±7.9	-
IWCV	81.0±2.3	79.6±2.2	82.0±5.8	84.4±6.2	84.0±7.0	81.2±2.9	78.8±8.0	77.4±7.6	82.0±5.8	85.2±5.5	-	84.2±6.8	-
DEV	81.6±7.7	78.2±2.7	86.4±3.9	83.2±5.2	84.9±6.3	78.8±6.6	82.0±6.5	77.3±4.0	81.7±6.8	83.2±5.2	-	82.8±6.8	-
DEV-N	82.0±7.2	80.7±3.1	81.1±7.3	84.1±6.5	83.7±3.4	83.2±5.8	80.7±5.7	80.2±6.8	82.8±5.7	85.5±5.9	-	78.3±9.7	-
Entropy	57.0±16	56.6±15	65.8±22	58.4±19	49.9±0.26	50.0±0.0	64.2±20	69.0±18	67.7±16	51.9±4.3	-	50.0±0.0	-
InfoMax	81.5±4.8	83.2±5.4	83.5±4.9	82.7±4.7	78.9±6.3	81.7±3.8	80.1±3.0	82.6±7.0	78.5±8.7	84.8±6.5	-	80.6±3.2	-
Corr-C	74.8±10	77.5±2.8	74.0±4.7	76.7±1.5	72.3±6.6	76.0±4.9	76.9±2.7	72.4±2.6	79.2±7.0	79.7±5.4	-	76.9±2.7	-
MCC (V)	83.8±6.7	82.7±5.3	86.7±6.4	85.6±5.8	84.6±6.8	84.7±6.3	82.4±6.4	78.6±7.2	79.9±7.6	74.4±19	-	82.8±3.9	-
BNM (V)	81.2±5.2	83.2±5.4	85.3±1.3	82.7±4.7	78.9±6.3	81.7±3.8	80.1±3.0	54.4±7.5	60.9±17	83.9±6.0	-	54.4±7.5	-
ClassAMI	81.6±6.9	82.2±4.8	82.0±5.4	82.4±4.4	83.4±7.4	81.5±6.3	83.0±5.1	79.1±6.1	67.7±9.9	84.8±4.3	-	81.6±5.6	-
SND	76.0±5.1	77.7±2.8	74.2±4.4	71.7±3.9	74.9±8.2	74.8±2.0	75.4±2.2	76.5±4.5	78.5±5.5	75.3±9.9	-	71.2±5.7	-
MixVal	83.8±6.1	63.9±13	66.5±16	75.2±13	70.6±16	75.6±13	65.9±18	75.8±15	65.9±17	71.0±18	-	59.0±12	-
TransScore	82.0±4.4	84.2±6.2	83.5±4.9	80.8±4.8	83.2±8.2	80.9±5.0	78.8±6.4	74.1±7.0	78.1±6.0	78.4±18	-	79.8±4.4	-

Table 12. Complete UDA results on ADNI-2→ADNI-3 (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 11. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	AD2A	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	85.9±4.5	89.3±5.0	92.0±3.9	90.8±4.1	90.3±3.7	90.3±5.6	88.9±3.7	90.7±3.4	87.5±3.6	89.1±3.5	89.9±1.3	92.1±3.1	90.7±6.5
Source-Risk	84.7±9.5	80.8±7.0	77.8±7.8	84.2±8.3	80.8±3.0	80.6±8.0	79.5±10	88.0±7.4	83.9±6.5	85.0±7.5	-	80.8±11	-
IWCV	81.1±2.9	80.8±2.7	79.6±6.1	84.3±7.7	80.1±8.1	81.9±3.8	77.2±10	88.0±6.0	80.7±9.0	80.6±6.9	-	81.9±8.9	-
DEV	80.9±11	77.5±3.3	88.1±4.7	83.2±6.6	84.8±7.7	78.7±7.8	80.0±7.3	80.4±6.0	77.5±5.5	84.2±6.9	-	79.6±8.2	-
Entropy	50.0±17	50.0±17	50.0±22	50.0±21	50.0±0.29	50.0±0.0	50.0±20	50.0±4.8	63.4±20	71.3±19	-	50.0±0.0	-
InfoMax	80.2±6.1	84.8±6.7	85.6±6.0	81.9±5.9	77.8±7.2	82.2±4.2	79.5±3.7	84.2±7.5	83.4±9.3	74.7±8.8	-	81.4±3.7	-
Corr-C	77.6±13	79.0±3.4	75.6±5.7	76.5±1.8	73.5±9.0	75.5±6.1	77.3±3.0	81.4±6.8	72.4±3.5	81.0±8.5	-	77.3±3.0	-
MCC (V)	81.8±7.9	82.6±6.8	88.7±8.2	83.0±7.0	84.4±9.2	84.2±8.3	81.4±7.9	80.2±25	79.8±9.7	76.8±8.4	-	81.4±4.3	-
BNM (V)	80.2±6.9	84.8±6.7	85.6±1.5	81.9±5.9	77.8±7.2	82.2±4.2	79.5±3.7	84.2±7.5	53.6±10	65.7±22	-	53.6±10	-
SND	76.1±7.0	77.5±3.8	76.7±5.2	71.1±5.1	76.9±11	75.0±2.3	76.6±2.6	80.0±13	77.3±5.9	80.0±7.3	-	72.8±7.5	-
ClassAMI	81.5±9.4	81.0±6.1	84.4±5.9	81.7±5.9	83.5±10	80.5±6.9	82.1±6.5	86.0±4.7	76.5±7.5	69.5±13	-	82.0±7.5	-
DEV-N	84.7±9.5	80.8±4.3	77.8±8.7	84.1±8.3	84.8±3.9	81.9±7.1	79.5±7.3	88.0±6.8	82.8±8.4	83.8±6.9	-	77.2±13	-
MixVal	82.1±7.7	56.9±15	59.5±19	77.4±17	73.1±19	79.5±16	58.8±18	69.8±20	78.9±19	56.7±20	-	56.7±15	-
TransScore	80.2±5.4	86.8±7.7	85.6±6.0	80.1±5.9	83.9±9.7	82.2±6.2	81.4±8.1	83.6±23	73.0±9.1	78.2±7.6	-	81.4±5.6	-

Table 13. Complete UDA results on ADNI-1+2→AIBL (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	92.2±2.3	91.6±2.0	92.8±1.1	93.4±1.7	93.1±2.0	91.8±1.6	91.4±1.9	88.0±6.0	91.9±1.7	93.1±2.6	91.9±1.5	93.8±2.1	81.9±3.3
Source-Risk	88.4±3.0	85.0±3.5	86.7±4.5	87.8±3.9	85.6±8.2	87.6±2.7	84.0±3.2	79.9±4.9	86.2±4.6	86.5±2.0	-	84.0±5.1	-
IWCV	86.8±3.3	83.2±4.6	83.8±4.7	84.3±6.0	84.2±8.0	86.4±4.5	83.3±3.2	82.3±4.2	83.2±3.9	87.8±1.9	-	84.4±3.3	-
DEV	87.3±2.7	84.2±5.8	83.4±5.2	82.2±12	89.2±1.7	85.4±4.8	87.4±2.2	82.5±4.5	82.1±4.3	79.5±5.0	-	89.0±1.7	-
DEV-N	87.6±4.2	85.0±3.5	85.0±5.0	88.9±3.5	85.6±8.2	88.2±2.3	84.0±3.2	79.9±4.9	84.6±4.8	88.8±3.7	-	82.7±5.0	-
Entropy	88.3±3.3	84.5±5.4	88.6±4.8	86.9±8.0	66.0±20	66.7±15	79.0±16	80.7±7.8	79.1±16	80.5±18	-	64.8±16	-
InfoMax	84.7±6.7	82.4±4.9	78.9±3.2	84.2±5.0	79.5±7.2	79.4±5.5	81.6±5.0	82.2±6.4	85.8±4.2	82.9±2.9	-	81.6±5.0	-
Corr-C	80.4±6.8	77.3±4.7	75.4±3.0	77.5±3.6	77.6±5.8	74.3±4.7	75.0±3.4	77.3±3.2	77.4±3.6	75.9±3.9	-	75.1±3.2	-
MCC (V)	88.3±3.8	86.0±3.0	84.7±5.5	89.7±3.4	86.3±4.6	80.6±11	83.8±2.9	81.3±6.2	84.8±3.1	88.8±3.8	-	84.1±4.8	-
BNM (V)	88.2±4.1	84.7±3.0	80.2±1.1	83.9±5.2	82.1±8.4	79.4±5.5	81.6±5.0	65.7±8.2	74.6±13	86.7±6.3	-	65.7±8.2	-
ClassAMI	84.9±7.1	68.7±19	83.9±5.5	84.0±5.1	86.0±5.6	85.9±2.4	84.5±2.3	82.8±5.4	74.2±13	84.8±2.6	-	83.3±3.4	-
SND	74.0±1.8	75.2±3.8	76.0±5.6	77.9±3.6	76.6±6.8	75.2±4.7	74.1±5.7	74.8±2.0	78.3±4.4	73.4±4.4	-	76.9±5.2	-
MixVal	88.8±4.0	59.4±12	82.0±7.5	84.1±8.7	77.8±12	76.5±16	81.2±6.1	79.4±5.5	76.9±14	67.9±13	-	70.9±16	-
TransScore	88.5±4.0	82.6±5.8	79.5±3.9	82.1±6.7	80.1±8.3	80.7±6.0	85.0±1.3	80.5±5.9	82.1±4.0	85.1±3.4	-	80.1±4.7	-

Table 14. Complete UDA results on ADNI-1+2→AIBL (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 13. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	AD2A	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	91.5±2.9	91.8±2.1	92.5±1.4	92.5±1.9	92.0±1.9	91.5±1.9	91.3±2.5	92.0±3.2	89.6±7.9	92.3±2.1	91.7±0.91	92.5±2.4	80.3±4.7
Source-Risk	86.9±3.6	86.5±3.5	87.4±5.5	88.6±5.3	86.6±9.7	87.4±3.0	85.3±3.9	86.5±2.4	78.7±6.0	87.1±5.5	-	85.1±6.6	-
IWCV	86.9±4.1	84.0±5.9	85.1±6.3	85.2±7.6	85.1±11	84.9±6.0	84.2±3.8	88.5±2.4	83.3±5.5	82.8±5.5	-	84.9±4.2	-
DEV	87.4±3.0	86.0±6.8	84.8±6.8	87.0±15	89.1±2.3	84.8±6.4	87.0±2.8	81.8±6.3	83.7±6.0	82.5±5.2	-	88.5±2.3	-
Entropy	89.9±3.4	86.4±6.7	90.0±5.5	88.7±10	53.9±22	61.5±18	85.3±20	85.1±22	83.7±10	86.8±19	-	61.5±18	-
InfoMax	85.8±9.3	84.4±6.2	79.7±4.2	83.8±6.6	80.5±9.0	81.6±6.7	83.5±5.7	82.3±3.7	84.3±7.2	86.8±5.2	-	83.5±5.7	-
Corr-C	76.6±7.4	78.9±6.2	76.6±3.6	77.1±4.0	79.1±7.4	74.0±6.0	75.5±4.7	75.3±5.3	77.8±3.2	78.7±4.5	-	76.0±4.1	-
MCC (V)	87.9±4.3	86.8±3.5	84.3±6.9	88.4±4.2	87.7±5.8	85.6±13	84.8±3.8	88.2±4.9	77.8±6.4	85.4±3.8	-	85.6±6.2	-
BNM (V)	86.3±4.6	83.5±3.4	79.9±1.4	83.8±6.6	84.9±11	81.6±6.7	83.5±5.7	86.9±7.0	62.1±9.4	79.7±16	-	62.1±9.4	-
SND	74.4±2.1	76.0±5.1	76.3±7.1	77.8±4.5	73.0±8.0	74.0±6.3	73.0±7.1	73.2±5.8	75.0±2.1	78.8±5.8	-	75.5±6.4	-
ClassAMI	84.1±9.1	78.9±20	85.6±7.2	85.5±6.6	83.9±6.6	85.5±3.2	84.4±2.8	84.4±3.5	84.4±6.3	77.7±18	-	83.7±3.9	-
DEV-N	86.9±5.6	86.5±3.5	82.8±6.1	88.7±4.3	86.6±9.7	87.7±3.0	85.3±3.9	87.7±4.8	78.7±6.0	83.4±5.7	-	81.8±6.6	-
MixVal	90.6±4.9	57.7±14	79.7±8.2	86.6±11	82.5±14	82.8±19	79.9±7.6	61.5±15	77.2±6.6	81.1±18	-	75.5±19	-
TransScore	87.4±4.6	82.3±7.5	79.7±5.2	82.5±8.2	80.5±11	83.5±7.1	84.9±1.7	86.7±4.3	77.8±6.9	80.6±4.8	-	78.9±5.0	-

Table 15. Complete UDA results on RSNA→Child CXR (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	82.3±2.1	83.1±1.8	82.8±2.0	84.7±1.6	83.5±1.4	87.1±1.3	84.0±1.1	81.5±2.0	84.0±1.2	89.7±0.40	84.5±2.3	89.7±0.40	92.7±1.1
Source-Risk	73.7±4.2	72.2±1.2	73.4±2.2	72.4±4.2	72.6±2.6	83.5±2.4	75.4±5.9	51.5±14	78.0±4.5	84.8±3.0	-	73.2±4.3	-
IWCV	73.3±4.3	72.3±4.6	74.1±0.93	74.0±3.0	76.6±4.5	76.4±6.3	74.5±4.3	54.3±12	76.9±3.7	86.2±1.5	-	76.4±6.3	-
DEV	73.3±4.9	73.1±9.2	72.1±4.4	72.1±3.4	71.7±7.0	69.1±12	68.9±17	54.6±15	69.5±9.3	75.6±4.4	-	72.6±12	-
DEV-N	71.8±6.0	70.8±3.5	71.7±2.2	70.9±2.7	74.1±6.1	82.9±2.3	74.8±2.5	66.5±9.7	73.6±1.6	85.1±3.5	-	75.8±3.3	-
Entropy	65.6±12	58.0±9.9	56.9±6.1	63.2±14	51.7±3.4	56.2±12	52.1±3.0	57.2±7.7	68.9±11	71.0±17	-	51.0±0.68	-
InfoMax	71.8±3.6	74.5±3.5	78.6±2.1	80.2±3.5	72.0±5.3	81.6±1.2	72.4±6.7	57.2±12	76.2±0.48	86.8±1.8	-	78.8±5.9	-
Corr-C	72.0±3.3	76.2±5.2	77.4±7.0	81.0±3.7	74.2±2.5	81.7±1.2	71.2±14	63.0±17	69.5±22	87.1±1.6	-	79.1±6.2	-
MCC (V)	70.0±7.1	71.9±1.1	74.5±1.2	78.0±4.9	69.5±5.1	73.8±14	64.5±16	54.9±11	74.6±2.6	85.0±3.3	-	73.8±14	-
BNM (V)	71.9±3.4	74.5±3.5	76.8±2.8	75.6±13	72.0±5.3	82.0±1.6	72.3±6.5	57.2±12	75.3±1.8	86.6±1.6	-	78.8±5.9	-
ClassAMI	55.5±2.9	65.7±10	78.3±2.6	77.8±1.1	58.9±11	82.8±3.2	74.4±8.5	68.5±6.2	75.5±5.7	82.1±7.8	-	74.2±8.7	-
SND	73.3±6.3	75.4±4.2	75.5±6.5	75.8±6.2	71.8±5.5	73.1±5.5	70.4±9.8	60.8±10	78.6±5.0	85.1±1.4	-	70.5±9.7	-
MixVal	72.7±12	75.4±9.4	74.7±11	79.7±2.8	71.7±12	70.2±12	68.8±9.8	51.5±9.8	73.5±1.4	87.9±1.7	-	48.5±6.0	-
TransScore	72.8±4.0	74.8±2.6	74.8±1.3	75.0±2.2	71.1±6.3	71.7±12	63.6±13	57.2±11	74.9±2.8	86.6±1.7	-	70.3±12	-

Table 16. Complete UDA results on RSNA→Child CXR (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 15. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	CoUDA	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	81.9±2.6	83.1±2.1	82.1±2.4	83.8±1.9	83.8±1.9	87.3±1.8	83.6±1.5	89.7±0.49	80.8±2.5	83.5±1.5	84.2±2.7	89.7±0.49	93.0±1.2
Source-Risk	73.4±5.6	72.3±1.7	74.4±2.8	71.0±5.4	72.4±3.7	84.4±2.9	75.0±8.0	84.9±4.0	52.3±15	77.5±6.0	-	72.4±6.0	-
IWCV	74.0±5.2	74.1±6.0	73.7±1.1	75.6±3.5	77.8±5.4	75.1±7.8	73.6±5.8	85.6±1.7	55.0±15	76.7±4.6	-	75.1±7.8	-
DEV	73.0±5.9	75.8±11	74.2±5.3	73.6±3.7	73.0±9.4	71.9±17	71.2±22	77.9±4.9	52.9±20	72.1±12	-	74.4±16	-
Entropy	68.8±12	51.1±9.9	56.7±6.9	63.6±17	50.0±3.9	51.2±14	51.1±3.6	79.7±18	55.6±7.7	72.1±13	-	51.1±0.92	-
InfoMax	71.4±4.9	74.7±4.3	78.5±2.8	80.8±4.9	75.2±5.8	81.7±1.5	70.5±8.6	85.9±2.0	62.6±15	76.1±0.51	-	81.2±7.2	-
Corr-C	71.8±4.5	79.5±5.2	79.5±8.3	80.6±4.7	75.2±3.0	81.8±1.5	72.8±17	86.3±1.6	64.6±23	80.0±26	-	81.7±7.3	-
MCC (V)	71.4±8.0	71.7±1.2	74.8±1.6	78.3±6.8	69.5±6.1	80.4±17	71.7±20	85.8±4.5	53.9±14	75.8±3.0	-	80.4±17	-
BNM (V)	71.4±4.6	74.7±4.3	77.6±3.5	80.8±17	75.2±5.8	82.0±2.1	70.5±8.3	85.9±1.8	62.6±15	75.8±2.4	-	81.2±7.2	-
SND	71.5±7.7	76.1±5.2	77.8±8.2	78.3±6.7	73.2±6.5	71.9±7.2	72.2±13	85.0±1.7	63.0±13	79.1±6.2	-	70.1±13	-
ClassAMI	56.8±3.4	62.6±13	77.8±3.5	77.8±1.5	53.6±14	84.2±3.9	76.9±9.6	83.9±10	65.3±7.6	75.8±7.6	-	78.2±9.7	-
DEV-N	73.4±6.4	72.2±4.2	72.9±2.5	70.4±3.5	72.0±7.2	83.1±3.1	74.9±3.1	87.0±4.0	68.4±13	73.0±1.8	-	75.4±4.0	-
MixVal	76.3±14	78.9±11	78.0±14	80.3±3.1	75.9±16	72.7±15	71.0±12	88.0±2.0	50.6±14	73.1±1.6	-	50.3±7.8	-
TransScore	73.3±5.6	74.7±3.5	74.8±1.4	74.5±2.7	68.8±8.3	76.0±15	68.5±15	87.7±1.9	62.6±14	75.8±3.4	-	72.3±15	-

Table 17. Complete UDA results on Child CXR→RSNA (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	75.1±0.36	74.6±0.99	77.2±1.1	76.2±0.32	78.1±0.45	77.3±0.66	77.2±0.47	76.3±0.89	77.0±0.39	73.2±0.44	76.4±1.5	78.3±0.26	82.4±0.46
Source-Risk	72.4±2.8	69.3±4.7	68.8±2.7	72.0±1.3	74.6±2.1	73.5±1.8	71.8±3.1	61.9±17	67.5±2.9	71.5±0.84	-	70.7±3.0	-
IWCV	73.5±1.5	71.7±3.8	70.5±3.0	70.2±1.2	76.1±1.4	70.1±7.5	69.3±4.5	59.2±12	70.9±3.5	71.9±0.64	-	58.7±11	-
DEV	73.5±1.5	66.6±5.8	69.3±3.7	68.2±4.6	75.7±1.0	74.3±2.4	67.7±9.6	53.5±18	67.9±6.6	71.9±0.64	-	63.3±18	-
DEV-N	71.7±2.8	68.4±4.4	68.8±2.7	71.5±1.2	74.9±2.1	73.5±1.8	71.8±3.1	59.4±19	67.5±2.9	71.5±0.84	-	70.6±3.6	-
Entropy	72.4±1.1	66.9±1.4	68.3±1.8	67.2±4.6	76.2±1.6	74.2±1.6	72.8±3.4	68.0±7.0	75.5±0.70	71.2±0.32	-	74.2±1.6	-
InfoMax	72.5±1.0	72.1±2.9	73.2±1.0	74.5±0.94	75.7±1.2	74.2±2.0	74.1±0.98	65.7±16	74.2±1.7	71.2±0.32	-	74.2±2.0	-
Corr-C	72.3±0.96	74.0±2.0	76.3±1.1	75.8±0.36	76.0±0.51	75.5±1.1	73.9±0.92	71.6±2.9	74.0±1.9	71.7±0.49	-	74.6±1.4	-
MCC (V)	72.6±1.2	67.5±2.1	68.8±1.8	69.1±3.4	74.3±4.3	74.8±2.1	74.4±1.4	72.5±2.8	75.7±0.99	71.2±0.32	-	72.0±1.3	-
BNM (V)	72.5±1.0	72.1±2.9	73.2±1.0	74.5±0.94	75.7±1.2	74.2±2.0	74.0±0.98	65.7±16	73.8±1.1	71.2±0.32	-	74.2±2.0	-
ClassAMI	71.4±1.2	69.2±1.8	69.5±1.7	70.9±2.5	75.3±0.84	73.8±1.4	72.9±1.5	70.4±3.2	70.0±3.7	71.3±0.36	-	72.9±2.3	-
SND	73.0±1.4	73.9±0.93	74.7±5.6	76.2±0.32	74.7±1.2	73.1±2.1	71.6±4.3	63.4±13	71.2±2.6	72.4±0.73	-	74.7±1.2	-
MixVal	72.6±1.1	68.0±2.3	70.3±2.6	71.7±3.3	76.7±0.72	73.7±0.94	73.2±1.1	61.6±16	73.3±1.1	71.2±0.28	-	73.3±1.1	-
TransScore	70.5±2.8	70.3±0.90	70.8±1.4	72.7±2.7	76.8±0.93	73.7±1.1	73.5±2.2	56.0±20	75.3±0.91	71.3±0.29	-	75.3±0.91	-

Table 18. Complete UDA results on Child CXR→RSNA (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 17. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	CoUDA	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	75.2±0.46	74.7±1.1	77.4±1.2	76.2±0.40	78.2±0.58	77.4±0.87	77.0±0.62	72.9±0.47	76.6±1.0	77.2±0.46	76.4±1.6	78.3±0.37	82.4±0.64
Source-Risk	73.5±3.4	72.2±5.1	70.1±3.0	72.5±1.3	73.7±2.5	73.5±2.5	73.3±3.8	71.9±1.0	71.4±20	67.4±3.5	-	71.3±3.9	-
IWCV	73.5±2.0	73.5±4.6	71.7±3.9	70.4±1.6	76.3±1.9	72.2±8.5	71.0±5.7	71.8±0.81	63.1±15	71.4±4.3	-	60.4±15	-
DEV	73.5±2.0	68.3±7.1	67.8±4.2	69.5±5.8	76.3±1.2	74.5±3.0	73.7±11	71.8±0.81	63.1±21	69.8±8.4	-	70.8±21	-
Entropy	72.5±1.3	66.2±1.7	69.2±2.0	65.6±5.5	76.8±1.8	74.1±2.1	73.2±4.2	71.2±0.42	69.2±8.5	75.5±0.82	-	74.1±2.1	-
InfoMax	72.6±1.3	71.1±3.2	73.2±1.3	74.8±1.2	74.9±1.3	74.0±2.5	74.2±1.3	71.2±0.42	73.2±18	74.1±2.3	-	74.0±2.5	-
Corr-C	71.7±1.1	74.7±2.4	76.7±1.3	75.6±0.36	75.9±0.61	76.0±1.3	74.1±1.1	72.0±0.57	71.8±4.0	73.4±2.4	-	74.9±1.9	-
MCC (V)	73.1±1.4	66.5±2.6	69.2±2.0	69.8±4.4	76.1±5.2	74.2±2.5	75.2±1.6	71.2±0.42	73.2±3.4	76.1±1.2	-	71.7±1.7	-
BNM (V)	72.6±1.3	71.1±3.2	73.2±1.3	74.8±1.2	74.9±1.3	74.0±2.5	74.1±1.3	71.2±0.42	73.2±18	74.1±1.5	-	74.0±2.5	-
SND	73.2±1.9	73.7±1.2	77.4±6.7	76.2±0.40	75.0±1.5	74.0±2.5	69.6±4.6	72.2±0.95	68.9±17	72.0±3.5	-	75.0±1.5	-
ClassAMI	71.3±1.4	69.2±2.4	70.3±2.1	71.5±3.1	75.2±1.1	73.5±1.8	72.1±1.8	71.2±0.43	71.8±3.5	70.6±4.4	-	73.0±2.4	-
DEV-N	70.5±3.4	67.9±5.1	70.1±3.0	70.6±1.3	75.9±2.2	73.5±2.5	73.3±3.8	71.9±1.0	71.4±21	67.4±3.5	-	72.4±4.0	-
MixVal	72.9±1.3	68.1±3.3	71.6±3.1	72.3±4.1	76.7±0.88	74.0±1.1	72.7±1.3	71.2±0.35	68.0±20	73.4±1.3	-	73.4±1.3	-

Table 19. Complete UDA results on LDD→CRD (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	79.0±0.82	82.1±0.52	82.4±0.54	82.6±0.42	83.0±0.52	82.4±0.90	83.2±0.50	83.0±0.50	83.0±0.63	81.7±0.80	82.6±0.47	83.6±0.13	95.7±0.34
Source-Risk	67.6±1.8	79.7±0.94	78.7±2.2	69.2±5.3	79.9±2.4	74.1±2.6	81.0±1.9	78.8±3.9	79.5±0.96	75.4±1.4	-	74.0±6.8	-
IWCV	69.5±4.2	79.5±0.43	77.7±3.4	71.8±3.6	79.2±2.1	74.7±2.2	79.7±3.1	75.4±6.5	73.8±5.4	76.1±1.1	-	74.7±2.2	-
DEV	73.4±6.4	77.9±3.3	79.8±0.45	77.6±2.1	79.8±3.8	72.9±8.8	76.2±4.7	54.6±12	79.8±3.2	79.3±1.6	-	74.8±10	-
DEV-N	68.2±2.0	79.9±0.72	78.2±1.3	69.8±5.2	80.1±2.5	74.1±2.6	81.4±1.1	78.8±3.9	79.2±1.3	75.5±1.3	-	73.9±6.8	-
Entropy	62.4±7.0	73.8±13	68.6±17	68.7±11	75.2±14	70.5±12	62.7±17	73.5±14	80.8±1.8	75.2±0.08	-	54.3±9.7	-
InfoMax	74.0±4.3	81.0±1.3	81.5±0.67	81.4±0.95	82.2±0.88	81.5±1.4	80.6±2.6	80.6±1.5	81.9±0.90	78.8±0.85	-	81.9±0.59	-
Corr-C	76.7±2.9	79.6±1.4	81.5±0.76	79.5±2.4	82.3±1.0	79.2±2.6	80.8±1.4	78.8±6.2	81.9±1.4	79.2±2.1	-	81.1±1.6	-
MCC (V)	67.4±0.93	79.8±0.76	79.9±1.6	74.4±1.3	81.3±1.6	75.7±5.2	81.7±0.49	79.9±3.2	81.1±1.9	75.2±0.08	-	75.7±5.2	-
BNM (V)	72.8±3.5	80.7±0.93	81.5±0.67	81.4±0.95	82.2±0.88	81.5±1.4	80.6±2.6	80.6±1.4	81.8±1.1	78.3±0.93	-	81.9±0.58	-
ClassAMI	69.5±7.8	76.8±2.7	80.8±1.2	72.5±6.2	79.3±2.6	78.5±3.9	79.2±1.9	80.7±1.2	76.6±4.2	77.2±1.6	-	78.6±3.2	-
SND	68.5±4.8	72.6±4.4	74.6±4.1	71.4±2.3	76.9±1.3	68.5±4.6	72.2±3.3	72.8±4.5	72.3±3.4	77.9±3.0	-	75.8±2.6	-
MixVal	69.3±8.8	80.7±0.95	80.9±0.85	74.6±5.2	80.7±1.6	75.9±6.5	78.5±1.6	71.2±22	79.0±4.2	76.6±0.63	-	79.0±4.2	-
TransScore	70.1±3.9	80.0±1.2	81.4±1.2	79.6±1.0	81.3±1.3	81.5±1.4	81.7±0.90	62.9±20	81.3±0.65	77.1±0.89	-	81.2±0.99	-

Table 20. Complete UDA results on LDD→CRD (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 19. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	CoUDA	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	79.1±1.1	82.2±0.62	82.3±0.66	82.5±0.58	83.3±0.55	82.5±1.2	83.4±0.64	82.1±0.87	82.8±0.56	82.8±0.73	82.7±0.46	83.6±0.15	96.0±0.36
Source-Risk	68.8±2.1	80.1±1.1	77.8±2.7	72.2±5.8	80.8±3.1	74.7±2.9	81.9±2.3	75.0±1.9	80.6±4.7	79.5±1.2	-	77.4±7.7	-
IWCV	70.6±4.7	79.7±0.56	77.8±4.5	72.5±4.8	80.0±2.6	76.0±2.7	81.1±3.8	75.4±1.2	77.7±8.2	74.7±7.3	-	76.0±2.7	-
DEV	72.8±7.9	79.3±4.0	79.8±0.61	77.9±2.6	81.3±4.5	75.9±11	77.6±6.3	79.8±2.0	48.3±12	80.8±3.8	-	80.0±12	-
Entropy	64.8±8.8	79.6±15	80.4±16	72.3±13	80.2±16	74.8±15	50.0±16	75.2±0.11	80.4±16	81.2±2.2	-	50.0±11	-
InfoMax	73.8±5.6	81.2±1.5	81.4±0.88	81.5±1.2	82.2±1.0	81.4±1.9	81.7±3.2	78.5±1.0	80.8±1.9	81.7±1.1	-	81.8±0.71	-
Corr-C	76.9±3.7	79.6±1.9	81.5±1.1	80.5±2.7	81.8±1.1	80.0±2.9	81.0±1.7	79.5±2.6	81.3±7.4	82.4±1.7	-	81.7±1.7	-
MCC (V)	67.8±1.1	79.6±0.98	80.4±2.0	74.7±1.7	80.7±1.7	77.8±6.4	81.6±0.63	75.2±0.11	81.0±3.9	82.0±2.2	-	77.8±6.4	-
BNM (V)	73.8±4.8	81.1±1.1	81.4±0.88	81.5±1.2	82.2±1.0	81.4±1.9	81.7±3.2	77.9±1.1	80.8±1.9	81.7±1.5	-	81.8±0.71	-
SND	68.5±6.1	74.7±4.6	76.4±5.0	70.9±3.1	76.6±1.8	69.5±5.7	71.5±4.0	78.1±3.9	72.8±5.5	70.8±4.2	-	76.3±3.6	-
ClassAMI	73.3±9.5	76.5±3.2	80.5±1.4	74.0±7.9	80.2±3.4	80.5±4.6	78.9±2.3	77.3±2.1	80.3±1.4	76.7±5.7	-	78.3±4.3	-
DEV-N	68.8±2.5	80.1±0.91	77.8±1.5	72.2±6.4	80.8±3.1	74.7±2.9	81.9±1.3	75.9±1.6	80.9±4.7	79.5±1.5	-	76.5±7.8	-
MixVal	68.2±11	80.7±1.3	81.4±0.91	74.3±6.5	80.9±2.0	77.8±7.5	79.1±2.1	77.0±0.73	81.0±25	80.6±5.1	-	80.6±5.1	-
TransScore	69.0±4.9	79.6±1.5	81.6±1.7	79.8±1.3	81.0±1.7	81.4±1.9	81.5±1.1	77.1±1.2	71.3±24	81.6±0.72	-	81.8±1.1	-

Table 21. Complete UDA results on CRD→LDD (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	77.2±3.0	78.6±1.6	77.9±5.8	81.0±1.5	77.7±1.9	78.5±3.6	80.0±3.6	74.8±3.4	76.3±5.0	63.9±1.3	76.5±4.8	82.7±1.1	98.7±0.17
Source-Risk	68.2±1.6	44.9±1.3	64.7±11	66.2±14	58.4±12	57.1±9.7	60.0±8.2	49.8±7.5	56.0±8.4	55.2±1.2	-	55.2±1.2	-
IWCV	67.7±7.3	46.1±3.6	61.7±15	70.0±10	56.5±6.4	59.2±11	58.3±11	62.0±17	57.8±8.0	54.5±1.9	-	63.7±6.1	-
DEV	65.9±13	45.3±5.9	65.7±12	63.0±7.3	62.0±11	63.9±11	55.7±8.1	48.3±15	58.5±8.2	52.4±2.3	-	59.9±9.7	-
DEV-N	68.2±1.6	51.5±13	67.7±13	63.5±15	56.4±12	57.1±9.7	59.3±9.2	74.1±4.3	56.0±8.4	55.2±1.2	-	55.2±1.2	-
Entropy	65.4±8.9	44.8±1.6	59.2±14	73.9±13	58.8±11	68.1±7.9	52.2±9.6	47.6±3.0	61.5±9.9	54.1±1.6	-	58.4±7.7	-
InfoMax	67.1±2.5	50.5±11	73.1±9.8	69.9±10	66.4±8.3	55.0±11	54.4±11	51.5±10	61.6±9.6	54.2±1.6	-	55.0±11	-
Corr-C	64.1±1.6	63.1±11	73.1±9.2	63.5±10	67.0±8.1	53.3±2.6	55.4±2.3	51.2±14	55.8±7.9	54.7±1.2	-	68.1±12	-
MCC (V)	68.0±2.1	44.8±1.6	59.8±15	70.0±12	62.6±11	63.0±14	55.1±12	48.6±4.6	49.4±3.7	54.2±1.6	-	63.0±14	-
BNM (V)	67.4±1.9	50.4±11	73.1±9.8	69.3±11	66.4±8.3	55.0±11	54.4±11	45.2±16	61.6±9.6	54.2±1.6	-	55.0±11	-
ClassAMI	68.2±3.9	50.7±11	66.2±16	66.7±8.0	62.4±11	52.7±9.3	58.9±6.6	47.1±14	64.2±1.8	50.7±3.6	-	73.1±8.4	-
SND	60.6±2.9	61.5±11	71.2±4.6	67.8±8.8	53.3±8.4	51.0±2.0	57.9±9.2	62.8±8.2	64.9±8.3	54.7±0.90	-	51.8±7.2	-
MixVal	67.7±1.8	47.7±9.4	71.7±11	64.6±11	61.8±13	63.0±13	54.2±15	37.7±6.5	67.2±2.4	56.0±2.2	-	67.2±2.4	-
TransScore	68.0±1.9	66.8±11	66.0±13	70.4±13	52.0±5.6	54.6±15	53.5±7.1	51.8±10	53.5±6.2	54.8±1.1	-	55.9±6.7	-

Table 22. Complete UDA results on CRD→LDD (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 21. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	CoUDA	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	76.7±4.1	77.7±1.9	80.1±7.0	81.4±1.9	78.2±2.3	76.5±3.8	80.0±4.6	64.1±1.7	75.5±4.0	76.1±6.6	76.6±5.1	82.7±1.4	98.8±0.20
Source-Risk	68.1±2.0	45.0±1.7	64.6±12	67.0±17	59.0±12	60.4±13	62.9±8.9	55.1±1.7	44.9±8.2	58.5±9.3	-	55.1±1.7	-
IWCV	65.4±9.5	45.7±4.5	52.9±15	70.9±13	59.5±7.9	62.4±14	64.7±12	54.4±2.5	70.8±18	58.0±11	-	64.7±8.3	-
DEV	71.9±15	43.3±6.1	71.5±14	61.4±8.4	58.6±12	62.4±14	54.0±8.6	52.8±3.1	51.8±20	62.9±9.7	-	58.6±12	-
Entropy	70.0±11	44.6±2.1	50.2±14	80.0±15	57.6±15	65.8±9.5	50.1±13	54.1±2.0	47.6±3.1	66.6±10	-	62.8±7.9	-
InfoMax	66.9±3.2	45.4±13	77.0±11	75.7±11	68.5±11	53.0±14	50.4±14	54.1±2.0	55.4±12	64.5±11	-	53.0±14	-
Corr-C	64.9±1.9	69.8±11	76.5±11	58.4±12	68.5±11	54.7±2.8	55.0±3.1	54.4±1.6	51.0±18	53.7±10	-	76.3±12	-
MCC (V)	68.1±2.3	44.6±2.1	50.2±15	73.8±13	62.4±14	62.8±19	51.9±13	54.1±2.0	47.6±5.5	47.6±4.4	-	62.8±19	-
BNM (V)	66.9±2.4	45.4±13	77.0±11	75.7±11	68.5±11	53.0±14	50.4±14	54.1±2.0	55.4±18	64.5±11	-	53.0±14	-
SND	60.0±3.4	59.8±14	72.2±5.6	69.2±11	52.6±10	51.0±2.8	53.0±11	55.0±1.0	65.3±10	64.7±11	-	52.6±10	-
ClassAMI	68.6±4.4	46.0±14	75.6±17	63.6±9.7	61.5±13	58.8±9.5	56.1±7.5	50.5±4.8	45.5±19	64.2±2.1	-	77.3±9.1	-
DEV-N	68.1±2.0	47.0±15	75.1±13	67.0±17	49.0±12	60.4±13	62.9±11	55.1±1.7	75.5±5.2	58.5±9.3	-	55.1±1.7	-
MixVal	67.4±2.1	43.4±11	75.9±13	61.8±13	68.3±13	65.4±18	50.1±20	56.3±2.9	36.8±8.0	66.3±2.6	-	66.3±2.6	-

Table 23. Complete UDA results on OCT→SLO (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	61.9±1.3	66.7±0.52	67.0±0.39	67.2±0.53	66.1±0.79	64.2±1.4	65.0±0.66	60.4±2.5	64.2±2.4	65.1±2.1	67.4±0.15	75.0±0.58
Source-Risk	59.5±1.1	63.3±1.6	66.3±0.59	64.0±0.65	63.8±1.2	57.6±3.9	55.8±5.0	38.1±4.1	54.2±3.5	-	57.6±13	-
IWCV	56.2±5.5	65.8±0.42	61.0±1.4	59.7±1.4	56.0±4.2	60.2±4.1	52.5±10	47.1±5.9	54.8±2.2	-	50.3±10	-
DEV	57.1±4.1	64.2±2.4	61.6±3.7	62.2±1.8	54.1±1.4	56.1±6.2	57.9±4.4	51.2±3.8	55.7±3.6	-	62.7±2.8	-
DEV-N	58.7±0.49	63.3±1.6	65.7±1.6	63.8±0.16	63.4±1.9	59.2±4.5	55.8±5.0	39.6±6.6	48.8±4.2	-	57.9±9.3	-
Entropy	58.0±3.3	63.2±1.5	66.1±0.53	63.6±1.6	64.0±0.60	50.4±3.9	50.1±0.71	45.4±4.6	48.8±4.8	-	50.2±3.9	-
InfoMax	59.2±1.2	63.7±1.9	65.8±0.23	66.4±0.58	64.0±0.60	54.8±7.9	56.4±6.4	44.6±3.9	49.5±4.2	-	49.7±4.0	-
Corr-C	58.9±1.9	63.8±2.0	65.8±1.1	65.9±0.74	64.5±0.21	59.7±3.0	54.5±5.1	41.5±4.7	53.5±4.3	-	57.3±7.0	-
MCC (V)	59.1±1.6	63.3±1.4	66.1±0.53	65.0±0.97	64.0±0.60	54.0±7.4	54.8±8.8	45.4±4.6	48.8±4.8	-	54.0±7.4	-
BNM (V)	59.2±1.2	63.4±1.6	66.2±0.43	66.4±0.58	64.0±0.60	52.2±5.3	56.4±6.4	44.6±3.9	49.5±4.2	-	49.7±4.0	-
ClassAMI	55.5±6.7	64.8±1.1	65.3±0.98	63.7±1.0	64.8±0.38	54.0±6.8	60.4±2.8	41.6±8.0	51.2±7.8	-	50.2±5.9	-
SND	58.9±1.8	62.7±1.0	62.5±0.94	61.6±2.9	59.4±0.87	60.8±3.8	57.0±10	55.3±8.5	60.3±8.0	-	59.4±0.87	-
MixVal	56.4±6.5	62.8±1.3	64.8±0.91	65.6±0.70	64.0±0.25	58.7±4.5	51.3±2.9	44.8±6.6	51.5±3.7	-	51.5±3.7	-
TransScore	59.2±1.2	64.4±0.54	65.7±1.1	65.2±0.94	64.2±0.20	56.8±4.2	47.3±2.3	46.3±4.0	54.5±3.8	-	54.5±3.8	-

Table 24. Complete UDA results on OCT→SLO (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 23. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	61.5±1.3	66.5±0.49	66.9±0.39	67.4±0.50	66.5±0.73	64.1±1.4	65.3±0.63	61.3±2.4	63.8±2.4	65.2±2.0	67.4±0.15	75.7±0.57
Source-Risk	59.1±1.0	62.9±1.6	66.5±0.54	63.6±0.56	63.7±1.2	55.5±3.4	58.4±4.5	37.2±4.0	55.4±3.4	-	63.6±12	-
IWCV	56.8±5.4	65.6±0.36	60.7±1.4	59.6±1.4	55.9±4.2	61.4±3.9	48.8±9.7	48.2±5.8	55.5±2.2	-	48.8±10	-
DEV	56.8±4.1	65.5±2.0	63.2±3.4	61.2±1.6	53.8±1.4	57.6±6.1	59.2±4.2	51.0±3.8	55.5±3.6	-	64.2±2.4	-
Entropy	59.7±3.0	63.9±1.4	65.9±0.49	64.1±1.5	64.0±0.60	50.6±3.9	50.0±0.70	42.9±4.1	46.7±4.4	-	50.0±3.9	-
InfoMax	59.4±1.2	64.3±1.8	65.9±0.23	66.1±0.52	64.0±0.60	56.2±7.8	58.6±6.1	43.5±3.8	48.8±4.2	-	48.8±3.9	-
Corr-C	59.8±1.8	64.8±1.8	66.2±1.0	66.0±0.73	64.4±0.20	61.3±2.6	54.8±5.1	41.5±4.7	54.4±4.2	-	61.3±6.1	-
MCC (V)	59.7±1.5	63.9±1.2	65.9±0.49	65.0±0.97	64.0±0.60	54.2±7.4	50.0±7.7	42.9±4.1	46.7±4.4	-	54.2±7.4	-
BNM (V)	59.4±1.2	64.2±1.5	66.1±0.41	66.1±0.52	64.0±0.60	54.1±5.0	58.6±6.1	43.5±3.8	48.8±4.2	-	48.8±3.9	-
SND	59.8±1.6	63.0±0.99	62.7±0.91	63.0±2.6	59.1±0.82	62.4±3.6	62.9±9.2	59.0±7.8	62.5±7.7	-	59.1±0.82	-
ClassAMI	58.7±6.1	65.0±1.1	65.1±0.97	63.4±0.97	64.7±0.37	56.9±6.3	58.8±2.5	37.2±7.1	47.9±7.2	-	47.9±5.6	-
DEV-N	58.7±0.49	62.9±1.6	66.5±1.4	63.8±0.15	63.7±1.9	57.9±4.4	58.4±4.5	37.2±6.3	50.2±4.0	-	62.9±8.2	-
MixVal	60.0±5.7	63.1±1.3	64.9±0.91	65.7±0.69	64.0±0.25	60.5±4.2	50.4±2.8	47.9±6.0	51.4±3.7	-	51.4±3.7	-
TransScore	59.6±1.2	64.3±0.53	66.0±1.1	65.5±0.90	64.3±0.20	55.9±4.1	46.0±2.0	44.5±3.6	55.5±3.7	-	55.5±3.7	-

Table 25. Complete UDA results on SLO→OCT (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	60.7±1.2	63.0±1.0	64.6±1.5	65.4±0.34	63.6±1.1	62.3±1.4	61.3±3.3	58.4±5.2	62.0±1.6	62.6±2.0	65.6±0.62	74.4±0.38
Source-Risk	57.9±1.4	60.7±0.83	61.1±1.6	63.3±1.4	59.6±1.3	53.5±4.6	49.7±4.2	45.1±0.56	54.6±6.7	-	55.2±6.7	-
IWCV	56.6±0.78	56.2±0.90	60.5±2.5	60.8±0.25	58.7±0.39	54.1±3.6	48.6±1.2	51.2±3.5	55.7±4.7	-	52.0±5.4	-
DEV	58.4±1.7	56.8±3.4	61.4±2.3	60.3±1.8	56.1±1.9	55.9±3.4	51.8±4.9	46.2±1.5	57.5±3.0	-	54.2±7.8	-
DEV-N	57.0±3.0	61.6±2.3	62.4±0.69	62.2±1.6	60.3±1.7	52.6±6.2	49.4±4.5	43.7±2.5	49.7±3.8	-	60.1±4.9	-
Entropy	57.1±1.3	49.6±4.8	60.5±1.7	61.9±1.4	61.4±1.1	45.6±3.2	50.3±0.90	44.9±4.0	51.1±6.0	-	45.9±3.6	-
InfoMax	56.9±1.6	49.7±4.8	60.1±2.3	61.4±2.6	61.6±1.1	45.3±4.9	48.3±1.6	45.2±3.9	52.1±5.8	-	45.3±4.9	-
Corr-C	57.9±1.9	50.1±4.4	61.4±2.5	62.6±2.1	61.7±1.4	55.0±5.4	49.3±1.9	47.0±4.6	51.8±5.3	-	49.3±1.9	-
MCC (V)	56.9±1.5	49.6±4.8	60.5±1.7	61.0±1.2	61.4±1.1	45.5±3.1	49.7±1.6	44.9±4.0	50.8±5.6	-	45.5±3.1	-
BNM (V)	56.9±1.5	49.7±4.8	60.5±1.7	61.4±2.6	61.6±1.1	45.3±4.9	48.5±1.9	45.2±3.9	52.2±5.9	-	45.3±4.9	-
ClassAMI	57.8±2.1	59.8±1.5	59.5±2.5	60.4±1.1	61.8±1.6	52.6±7.5	47.0±1.4	43.6±3.8	52.7±6.0	-	46.0±5.9	-
SND	59.0±2.4	58.5±3.2	60.6±2.3	60.1±2.3	57.0±1.4	57.6±4.4	54.2±6.7	54.4±5.9	58.7±2.1	-	57.0±1.4	-
MixVal	58.1±1.9	49.8±4.4	59.7±2.1	62.3±2.2	60.0±1.4	46.2±4.6	50.7±2.7	44.8±3.5	53.6±5.6	-	52.2±6.5	-
TransScore	57.6±1.7	59.5±3.1	60.5±1.7	64.0±1.2	60.3±0.93	46.3±2.6	51.5±3.6	47.3±6.5	57.3±3.2	-	57.3±3.2	-

Table 26. Complete UDA results on SLO→OCT (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 25. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	61.2±1.1	62.4±0.89	64.0±1.4	65.5±0.33	63.3±1.1	63.0±1.2	62.8±3.0	60.2±4.9	61.2±1.4	62.8±1.6	65.5±0.62	75.8±0.30
Source-Risk	58.1±1.4	61.1±0.76	61.6±1.5	62.7±1.3	59.3±1.3	53.1±4.6	48.8±4.1	45.2±0.54	53.2±6.6	-	54.2±6.7	-
IWCV	56.2±0.73	56.6±0.81	60.6±2.5	60.9±0.23	58.8±0.37	53.1±3.5	48.2±1.1	50.3±3.4	57.4±4.5	-	50.3±5.2	-
DEV	58.3±1.7	56.9±3.4	60.1±2.0	60.7±1.8	56.2±1.9	56.8±3.3	49.9±4.6	45.3±1.3	59.2±2.6	-	58.2±6.9	-
Entropy	56.4±1.2	48.4±4.7	60.6±1.7	62.4±1.4	60.9±1.1	44.2±3.0	50.0±0.85	46.6±3.7	49.8±5.9	-	44.2±3.3	-
InfoMax	56.2±1.5	48.5±4.7	60.6±2.3	60.4±2.4	61.5±1.1	44.5±4.8	47.5±1.4	47.1±3.6	54.5±5.4	-	44.5±4.8	-
Corr-C	58.6±1.8	48.5±4.2	60.5±2.4	62.4±2.1	60.9±1.3	53.5±5.2	48.5±1.8	44.5±4.0	53.2±5.2	-	48.5±1.8	-
MCC (V)	56.2±1.4	48.4±4.7	60.6±1.7	60.3±1.0	60.9±1.1	44.2±2.9	49.6±1.6	46.6±3.7	49.8±5.6	-	44.2±2.9	-
BNM (V)	56.2±1.4	48.5±4.7	60.6±1.7	60.4±2.4	61.5±1.1	44.5±4.8	47.5±1.7	47.1±3.6	54.5±5.5	-	44.5±4.8	-
SND	59.3±2.4	58.9±3.2	60.0±2.3	60.5±2.3	56.8±1.4	58.5±4.3	56.5±6.5	56.0±5.7	59.8±1.9	-	56.8±1.4	-
ClassAMI	58.7±2.0	60.4±1.4	58.3±2.2	59.8±0.94	61.0±1.4	52.2±7.5	46.9±1.3	42.5±3.6	50.4±5.6	-	45.2±5.9	-
DEV-N	58.1±2.8	61.1±2.2	62.4±0.68	62.9±1.5	61.1±1.5	53.7±6.2	47.7±4.3	44.5±2.5	50.1±3.7	-	62.4±4.4	-
MixVal	58.6±1.8	47.9±4.1	59.7±2.1	61.7±2.2	60.7±1.3	48.2±4.3	50.9±2.7	45.3±3.5	52.4±5.5	-	48.8±5.8	-
TransScore	58.0±1.7	59.5±3.1	60.6±1.7	64.5±1.1	60.7±0.87	47.3±2.5	51.9±3.6	44.8±6.2	56.5±3.2	-	56.5±3.2	-

Table 27. Within-algorithm Spearman's ρ on ADNI-1→ADNI-2. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A
Source-Risk	0.80	0.76	0.74	0.74	0.74	0.70	0.53	0.79	0.80
IWCV	0.49	0.29	0.31	-0.28	0.29	0.04	0.05	0.48	0.38
DEV	0.56	0.49	0.47	0.04	0.24	0.21	0.19	0.57	0.57
DEV-N	0.77	0.74	0.72	0.71	0.73	0.69	0.53	0.78	0.79
Entropy	0.45	0.61	0.60	0.57	0.55	0.51	0.33	0.72	0.67
InfoMax	0.78	0.87	0.87	0.87	0.83	0.79	0.61	0.81	0.89
Corr-C	0.82	0.79	0.81	0.83	0.78	0.74	0.62	0.82	0.87
MCC (V)	0.70	0.83	0.82	0.81	0.76	0.73	0.55	0.78	0.86
BNM (V)	0.75	0.86	0.86	0.85	0.82	0.79	0.60	0.80	0.88
SND	0.75	0.58	0.57	0.65	0.50	0.45	0.38	0.78	0.66
ClassAMI	0.29	0.70	0.70	0.73	0.69	0.67	0.53	0.47	0.71
MixVal	0.22	0.33	0.37	0.33	0.35	0.35	0.25	0.31	0.29
TransScore	0.74	0.86	0.85	0.84	0.80	0.77	0.59	0.78	0.88

Table 28. Within-algorithm Spearman's ρ on ADNI-1→ADNI-3. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A
Source-Risk	0.52	0.52	0.61	0.40	0.47	0.40	0.48	0.72	0.75
IWCV	0.16	0.14	0.22	-0.12	0.15	0.11	0.13	0.50	0.36
DEV	0.22	0.26	0.35	0.11	0.18	0.22	0.03	0.49	0.48
DEV-N	0.51	0.52	0.61	0.39	0.48	0.40	0.28	0.70	0.75
Entropy	0.35	0.53	0.44	0.22	0.20	0.24	0.05	0.55	0.56
InfoMax	0.46	0.61	0.58	0.39	0.39	0.18	0.43	0.66	0.73
Corr-C	0.40	0.35	0.37	0.21	0.29	0.03	0.20	0.64	0.64
MCC (V)	0.45	0.66	0.64	0.41	0.40	0.30	0.42	0.64	0.70
BNM (V)	0.46	0.63	0.61	0.40	0.39	0.20	0.42	0.64	0.72
SND	0.34	0.14	0.24	0.18	0.24	-0.01	0.24	0.62	0.52
ClassAMI	0.02	0.56	0.58	0.37	0.46	0.41	0.44	0.41	0.61
MixVal	0.13	0.20	0.27	0.13	0.20	0.19	0.26	0.27	0.29
TransScore	0.42	0.60	0.59	0.37	0.38	0.16	0.41	0.55	0.71

Table 29. Within-algorithm Spearman's ρ on ADNI-2→ADNI-1. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A
Source-Risk	0.73	0.58	0.53	0.59	0.58	0.56	0.68	0.76	0.79
IWCV	0.37	-0.04	0.10	-0.28	0.39	0.16	0.03	0.45	0.30
DEV	0.40	0.32	0.33	0.17	0.30	0.22	0.11	0.47	0.47
DEV-N	0.70	0.61	0.58	0.62	0.61	0.60	0.68	0.76	0.81
Entropy	0.32	0.51	0.44	0.45	0.34	0.26	0.38	0.67	0.61
InfoMax	0.77	0.76	0.69	0.78	0.70	0.69	0.79	0.79	0.88
Corr-C	0.79	0.73	0.60	0.71	0.66	0.67	0.72	0.82	0.85
MCC (V)	0.69	0.71	0.66	0.74	0.56	0.57	0.76	0.77	0.86
BNM (V)	0.74	0.75	0.69	0.77	0.70	0.68	0.79	0.78	0.87
SND	0.63	0.31	0.18	0.37	0.42	0.49	0.38	0.71	0.61
ClassAMI	0.31	0.68	0.59	0.67	0.62	0.63	0.62	0.58	0.75
MixVal	0.24	0.25	0.25	0.36	0.29	0.26	0.55	0.30	0.35
TransScore	0.75	0.75	0.68	0.77	0.69	0.65	0.75	0.77	0.87

Table 30. Within-algorithm Spearman's ρ on ADNI-2→ADNI-3. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A
Source-Risk	0.72	0.68	0.66	0.69	0.67	0.57	0.57	0.74	0.82
IWCV	0.38	0.30	0.23	-0.23	0.20	0.01	0.24	0.54	0.60
DEV	0.43	0.48	0.36	0.18	0.36	0.29	0.36	0.68	0.68
DEV-N	0.64	0.61	0.57	0.58	0.58	0.49	0.52	0.71	0.78
Entropy	0.30	0.20	0.15	0.17	0.19	0.19	0.19	0.63	0.49
InfoMax	0.76	0.71	0.68	0.68	0.64	0.51	0.36	0.63	0.80
Corr-C	0.63	0.51	0.55	0.56	0.48	0.33	0.19	0.61	0.58
MCC (V)	0.70	0.64	0.60	0.62	0.60	0.57	0.41	0.68	0.73
BNM (V)	0.75	0.72	0.68	0.69	0.65	0.54	-0.04	0.55	0.78
SND	0.53	0.39	0.45	0.49	0.35	0.24	0.17	0.52	0.44
ClassAMI	-0.01	0.66	0.57	0.56	0.65	0.60	0.55	0.23	0.55
MixVal	0.37	0.29	0.27	0.32	0.30	0.23	0.24	0.28	0.31
TransScore	0.75	0.72	0.69	0.67	0.64	0.53	0.39	0.64	0.78

Table 31. Within-algorithm Spearman's ρ on ADNI-1+2→AIBL. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A
Source-Risk	0.71	0.56	0.65	0.56	0.62	0.54	0.58	0.66	0.77
IWCV	0.52	0.15	0.18	-0.28	0.25	0.08	0.22	0.41	0.53
DEV	0.60	0.45	0.37	0.21	0.43	0.30	0.43	0.51	0.67
DEV-N	0.71	0.56	0.62	0.56	0.62	0.56	0.57	0.67	0.76
Entropy	0.55	0.46	0.46	0.41	0.50	0.41	0.43	0.72	0.69
InfoMax	0.65	0.40	0.58	0.46	0.45	0.31	0.54	0.75	0.71
Corr-C	0.42	-0.04	0.24	0.06	0.04	-0.12	-0.09	0.41	0.35
MCC (V)	0.65	0.54	0.62	0.54	0.58	0.44	0.48	0.72	0.76
BNM (V)	0.66	0.47	0.61	0.50	0.50	0.35	-0.13	0.51	0.73
SND	0.34	-0.14	0.09	0.01	-0.11	-0.20	-0.18	0.24	0.16
ClassAMI	-0.25	0.37	0.38	0.33	0.46	0.50	0.39	0.29	0.60
MixVal	0.31	0.27	0.21	0.13	0.16	0.13	0.18	0.24	0.27
TransScore	0.60	0.42	0.57	0.44	0.42	0.29	0.34	0.64	0.69

Table 32. Within-algorithm Spearman's ρ on RSNA→Child CXR. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA
Source-Risk	0.11	0.20	-0.08	0.21	0.49	0.22	-0.10	0.47	0.57
IWCV	-0.05	0.14	-0.06	-0.02	0.41	0.17	0.03	0.30	0.42
DEV	-0.05	0.06	-0.13	-0.08	0.26	0.10	0.11	-0.01	0.24
DEV-N	-0.20	-0.17	-0.30	0.23	0.39	0.22	-0.08	0.17	0.46
Entropy	-0.16	-0.15	-0.39	-0.19	0.33	0.07	-0.14	0.26	0.34
InfoMax	0.29	0.32	0.01	0.09	0.65	0.58	0.01	0.52	0.56
Corr-C	0.75	0.75	0.63	0.18	0.58	0.59	0.07	0.70	0.65
MCC (V)	0.10	0.11	-0.22	-0.08	0.45	0.23	-0.14	0.33	0.47
BNM (V)	0.24	0.27	-0.04	0.08	0.65	0.57	0.01	0.49	0.55
SND	0.76	0.76	0.68	0.24	0.39	0.50	0.16	0.62	0.59
ClassAMI	-0.18	0.64	0.44	-0.28	0.57	0.44	0.04	0.07	-0.25
MixVal	0.43	0.37	0.46	0.36	0.30	0.12	-0.10	0.37	0.15
TransScore	0.05	0.09	-0.15	0.13	0.63	0.58	-0.02	0.24	0.54

Table 33. Within-algorithm Spearman's ρ on Child CXR→RSNA. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA
Source-Risk	0.19	0.11	-0.08	0.22	0.12	0.17	-0.08	0.02	-0.11
IWCV	0.20	0.16	0.12	0.14	0.23	0.09	-0.00	0.14	0.24
DEV	0.18	0.17	0.11	0.15	0.24	0.10	-0.02	0.11	0.23
DEV-N	0.19	0.10	-0.07	0.22	0.12	0.18	-0.08	0.03	-0.10
Entropy	-0.12	-0.29	-0.25	0.44	0.07	0.32	-0.01	0.60	-0.43
InfoMax	0.81	0.79	0.87	0.47	0.58	0.44	0.09	0.73	-0.36
Corr-C	0.96	0.97	0.99	0.43	0.61	0.38	0.13	0.69	-0.16
MCC (V)	-0.01	-0.15	0.04	0.42	0.15	0.34	-0.01	0.61	-0.42
BNM (V)	0.76	0.72	0.84	0.47	0.57	0.44	0.09	0.73	-0.37
SND	0.90	0.92	0.96	-0.27	0.58	0.29	0.15	0.46	0.58
ClassAMI	0.35	-0.08	0.16	0.21	0.05	0.12	0.12	0.45	0.06
MixVal	0.07	0.24	0.44	0.39	-0.19	-0.16	0.11	0.18	-0.14
TransScore	0.22	-0.05	-0.08	0.18	0.40	0.39	0.06	0.67	-0.37

Table 34. Within-algorithm Spearman's ρ on LDD→CRD. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA
Source-Risk	0.57	0.58	0.06	0.68	0.20	0.74	0.30	0.63	-0.08
IWCV	0.43	0.22	-0.19	-0.03	-0.18	-0.25	-0.02	-0.16	-0.16
DEV	0.43	0.28	-0.10	0.14	-0.09	-0.11	0.02	0.13	-0.09
DEV-N	0.55	0.55	0.04	0.66	0.18	0.72	0.28	0.57	-0.10
Entropy	0.53	0.57	-0.06	0.68	0.07	0.67	0.26	0.69	-0.07
InfoMax	0.78	0.85	0.85	0.87	0.89	0.84	0.39	0.93	0.40
Corr-C	0.65	0.75	0.87	0.81	0.87	0.76	0.39	0.83	0.84
MCC (V)	0.63	0.66	0.08	0.78	0.20	0.77	0.34	0.76	0.05
BNM (V)	0.77	0.84	0.81	0.86	0.87	0.84	0.39	0.93	0.35
SND	0.16	0.34	0.70	0.09	0.74	0.34	0.11	0.44	0.67
ClassAMI	0.30	0.15	0.23	0.36	0.26	0.68	0.45	0.48	0.38
MixVal	0.66	0.58	0.43	0.68	0.23	-0.08	0.24	0.42	0.42
TransScore	0.48	0.58	0.20	0.68	0.73	0.80	0.26	0.74	0.03

Table 35. Within-algorithm Spearman's ρ on CRD→LDD. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA
Source-Risk	-0.01	0.44	0.42	0.11	0.46	0.15	-0.04	-0.18	0.25
IWCV	0.28	0.13	0.19	0.09	0.10	0.13	0.15	0.17	0.20
DEV	0.22	0.17	0.24	0.17	0.16	0.12	-0.15	0.15	0.20
DEV-N	0.00	0.45	0.43	0.12	0.47	0.16	0.96	-0.18	0.26
Entropy	-0.21	0.34	0.28	-0.02	0.46	0.06	-0.07	-0.23	0.20
InfoMax	-0.02	0.49	0.47	0.15	0.03	0.31	0.01	-0.34	0.20
Corr-C	0.39	0.58	0.43	0.38	-0.13	0.35	0.05	-0.42	0.18
MCC (V)	-0.14	0.39	0.34	0.05	0.47	0.10	-0.05	-0.24	0.19
BNM (V)	-0.05	0.48	0.45	0.14	0.06	0.29	-0.02	-0.33	0.20
SND	0.50	0.21	0.15	0.37	-0.25	0.32	0.12	-0.14	-0.21
ClassAMI	-0.10	0.44	0.34	0.19	0.32	0.17	-0.18	-0.15	-0.08
MixVal	-0.12	0.32	0.29	0.06	0.38	-0.02	-0.18	0.01	0.18
TransScore	-0.14	0.30	0.28	0.04	0.12	0.24	0.02	-0.24	0.16

Table 36. Within-algorithm Spearman's ρ on OCT→SLO. Each cell is the mean ρ over three random seeds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD
Source-Risk	-0.08	0.40	0.27	0.59	-0.15	-0.12	-0.29	0.15
IWCV	0.06	-0.10	-0.06	-0.61	0.22	-0.02	0.02	-0.04
DEV	0.02	-0.06	0.02	-0.41	0.16	-0.09	0.07	-0.13
DEV-N	-0.05	0.42	0.21	0.58	-0.16	-0.12	-0.27	0.07
Entropy	-0.15	0.44	0.22	0.76	-0.11	-0.16	-0.34	0.15
InfoMax	-0.11	0.48	0.60	0.78	0.45	-0.04	-0.37	0.32
Corr-C	0.15	0.46	0.70	0.78	0.52	0.05	-0.26	0.37
MCC (V)	-0.14	0.45	0.26	0.77	-0.02	-0.14	-0.35	0.18
BNM (V)	-0.11	0.48	0.56	0.78	0.44	-0.05	-0.37	0.32
SND	0.26	-0.12	0.14	0.13	0.51	0.22	-0.02	0.34
ClassAMI	0.12	0.44	0.29	0.26	0.18	0.07	-0.06	-0.10
MixVal	-0.13	0.48	0.26	0.72	0.27	-0.07	-0.28	0.22
TransScore	-0.16	0.32	0.16	0.64	0.19	-0.16	-0.35	0.32

Table 37. Within-algorithm Spearman's ρ on SLO→OCT. Each cell is the mean ρ over three random seeds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD
Source-Risk	0.16	0.36	0.49	0.53	-0.35	-0.09	-0.26	0.19
IWCV	0.01	-0.06	-0.03	-0.40	0.15	0.02	0.08	0.04
DEV	-0.11	0.19	0.10	-0.00	0.01	0.07	0.05	-0.04
DEV-N	0.12	0.33	0.37	0.54	-0.34	-0.07	-0.26	0.18
Entropy	0.04	0.35	0.51	0.65	-0.39	-0.18	-0.44	0.13
InfoMax	0.09	0.36	0.59	0.68	-0.18	-0.07	-0.43	0.18
Corr-C	0.07	0.31	0.41	0.69	-0.06	-0.05	-0.33	0.26
MCC (V)	0.08	0.36	0.53	0.67	-0.40	-0.15	-0.44	0.14
BNM (V)	0.08	0.36	0.59	0.68	-0.19	-0.07	-0.43	0.18
SND	0.07	-0.02	-0.07	0.12	0.08	0.06	0.01	0.18
ClassAMI	-0.36	0.41	0.47	0.60	-0.23	-0.18	-0.26	-0.03
MixVal	0.08	0.35	0.50	0.61	-0.19	-0.13	-0.19	0.34
TransScore	0.05	0.31	0.48	0.17	-0.32	-0.07	-0.34	0.24